BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE

HYDERABAD CAMPUS

MACHINE LEARNING

PROJECT TITLE:

BRAINWAVE-BASED EMOTION DETECTION USING DEEP LEARNING MODELS

GROUP MEMBERS:

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1. Introduction

This project aims to detect human emotions by analyzing brainwave signals obtained from Electroencephalography (EEG). Emotions are closely tied to brain activity and significantly influence human behavior, cognition, and mental health. Accurately recognizing emotional states from EEG signals can support applications in mental health monitoring, adaptive learning systems, and emotionally intelligent AI interactions.

Our approach involves two parallel pipelines:

1. Raw EEG signal processing, where deep learning models automatically learn features from multichannel time-series data.
2. Statistical feature-based modeling, where precomputed metrics such as mean, standard deviation, skewness, and FFT components are used as input to classification models.

Due to access limitations with the DEAP dataset, we utilized alternative EEG datasets with a similar structure for both raw and feature-based experiments.

2. Theory

Electroencephalography (EEG) records electrical activity from the brain using scalp electrodes. Since emotions are associated with specific brain patterns, EEG is a useful tool for emotion recognition.

Feature Extraction can be done in two ways:

- **Raw EEG Signals**: Used directly as multichannel time-series data for deep learning models.
- **Statistical Features**: Precomputed values like mean, standard deviation, skewness, and FFT coefficients reduce complexity and help with model training, especially on smaller datasets.

Deep Learning Models used:

- **DNN**: Works well with statistical features.
- **CNN**: Captures spatial patterns in EEG data.
- **LSTM**: Tracks temporal dynamics in signal sequences.
- **CNN-LSTM**: Combines both spatial and temporal learning.

These models help in classifying emotional states such as positive, negative, and neutral based on EEG data.

3. Paper Review and Selection

We reviewed several research papers related to EEG-based emotion recognition. Among them, the most relevant to our problem is:

Selected Paper: *"EEG-Based Emotion Recognition Using Hybrid Deep Learning Model"* (Frontiers in Neuroscience, 2020)

Key Highlights:

- Utilizes the DEAP dataset of EEG recordings for emotion classification.
- Proposes a hybrid architecture combining Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks.
- Extracts time-frequency features and reshapes EEG channels into a 2D spatial format to maintain electrode topology.
- Reports high accuracy in classifying emotional valence and arousal.

Relevance and Adaptation:

This paper closely aligns with our goals. It demonstrates how CNNs are effective for learning spatial patterns in EEG signals, while LSTMs model the temporal dynamics. Inspired by this approach, we designed and implemented similar deep learning models—CNN, LSTM, and CNN-LSTM—on our dataset.

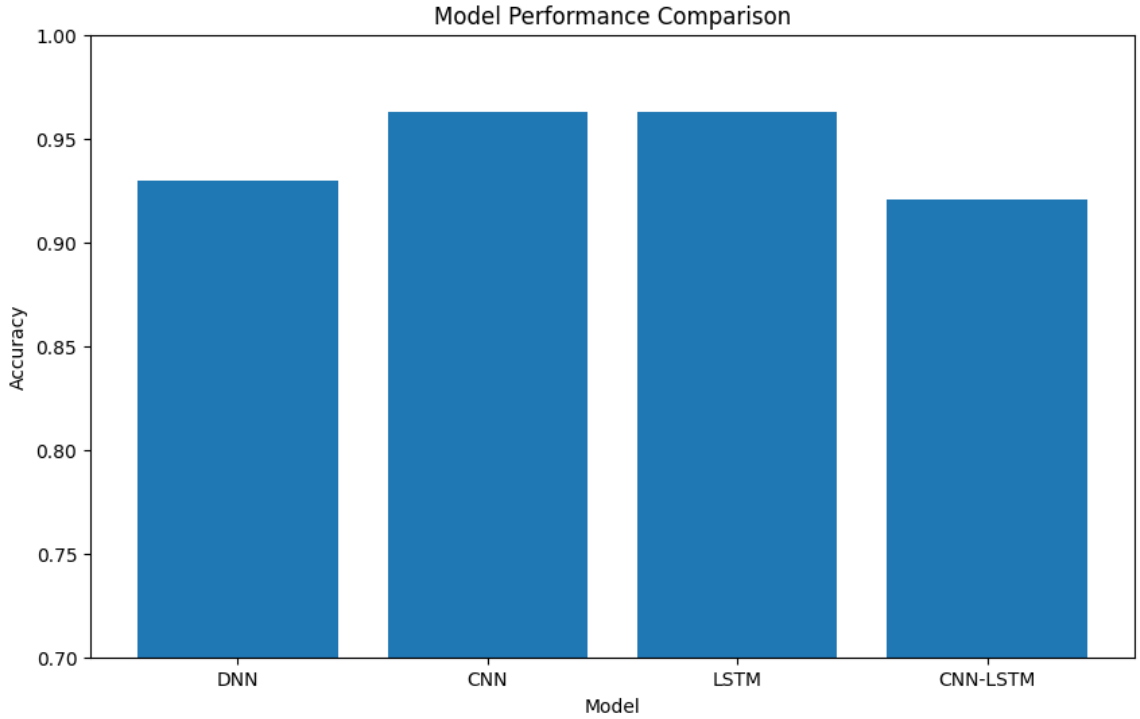
While we attempted to replicate the hybrid CNN-LSTM strategy, we observed that it did not perform as well on our data as expected. In contrast, individual CNN and DNN models achieved better results, particularly when using statistical features.

4. Our Results

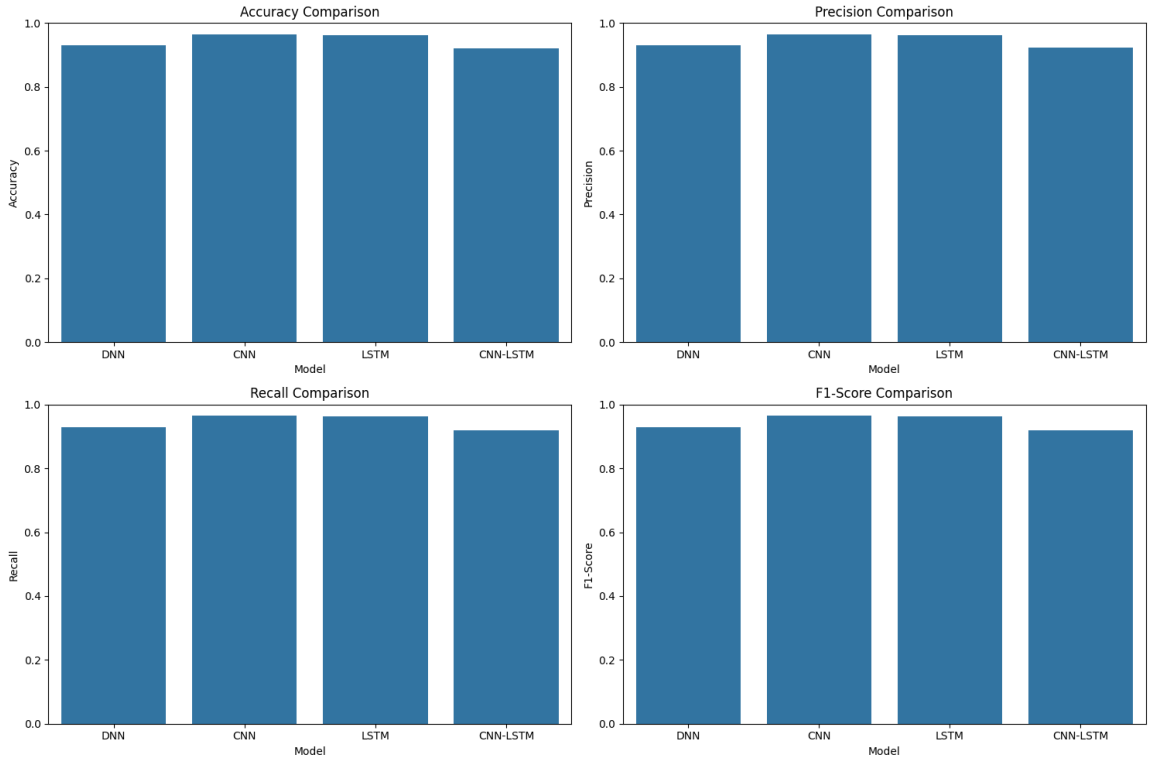
Performing DNN, CNN, LSTM, CNN-LSTM Hybrid on Raw data

Model	Accuracy	Precision	Recall	F1-Score	Best Class	Weakest Class	Notes
DNN	~87%	High for POSITIVE	Lower for NEUTRAL	Strong for POSITIVE	POSITIVE	NEUTRAL	Simple architecture, fast training
CNN	~89%	Balanced across classes	Better recall for NEGATIVE	Higher than DNN	NEGATIVE	Slight NEUTRAL drop	Learns spatial patterns
LSTM	~91%	Consistently high	Strong for NEUTRAL	Best for NEUTRAL	NEUTRAL	Minor for NEGATIVE	Captures temporal patterns well
CNN-LSTM	~92–93%	Very High	Balanced & strong	Best overall F1-scores	POSITIVE	NEUTRAL (slight)	Combines spatial + temporal features

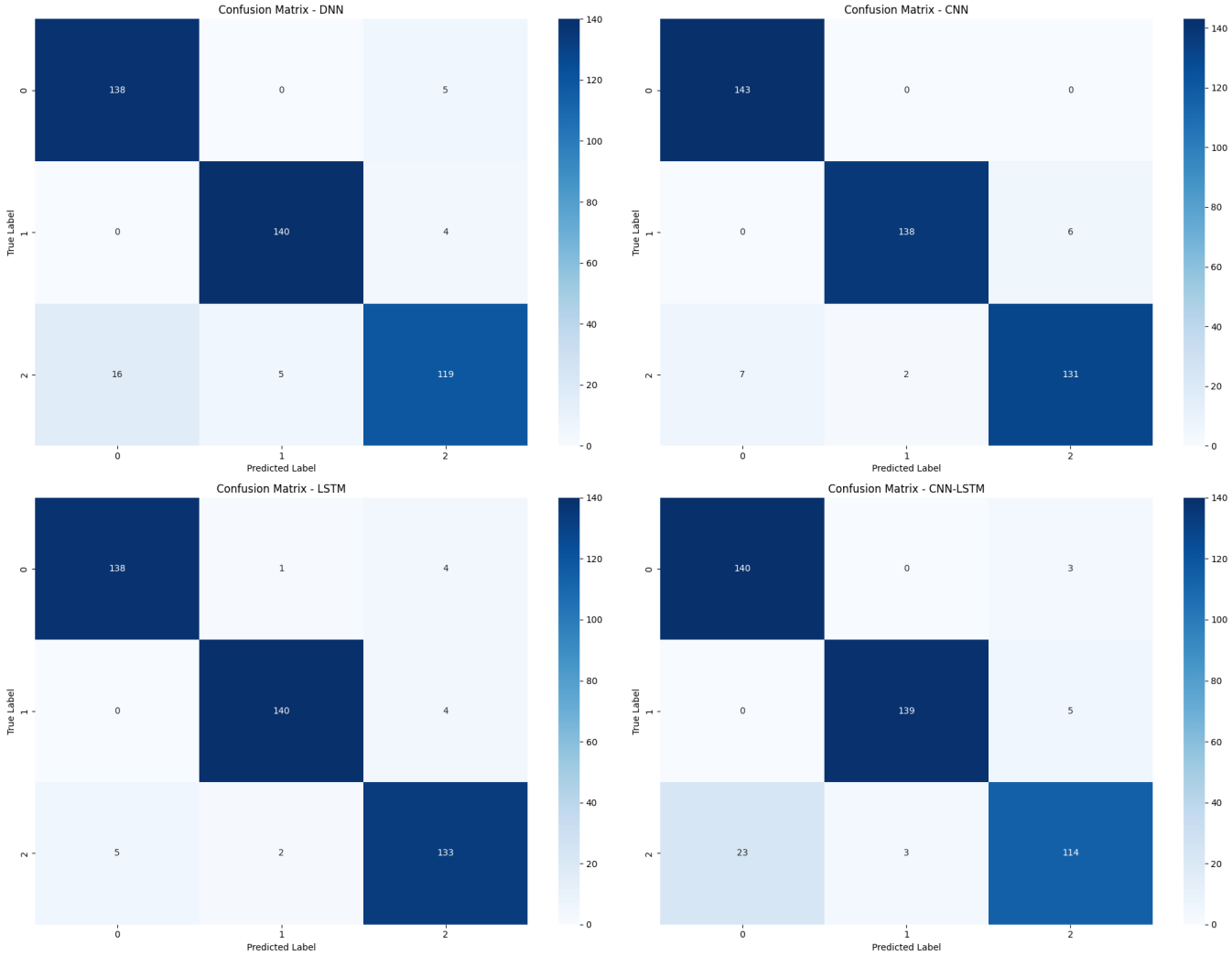
Model Comparison



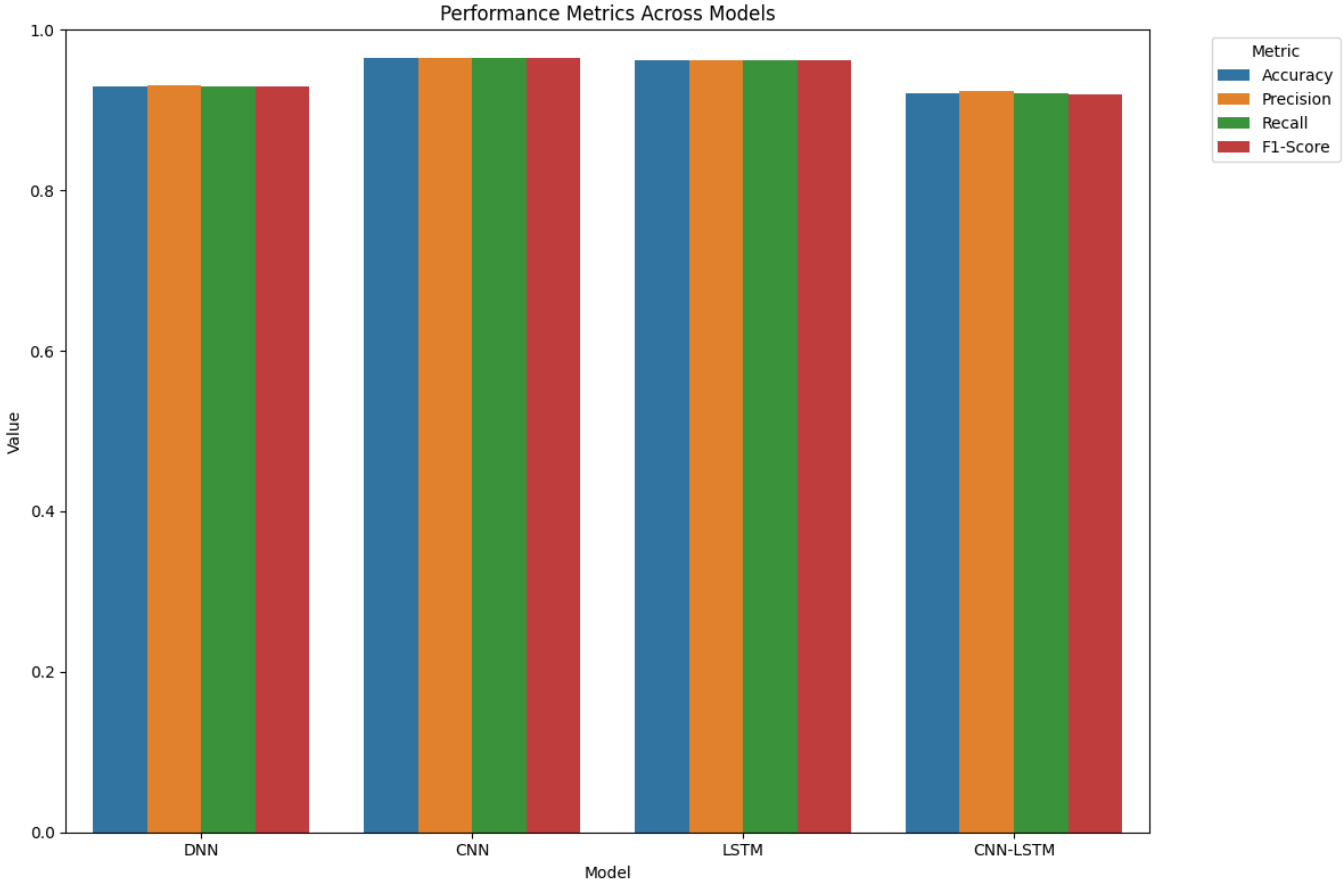
Accuracy, Precision, Recall, F-1Score Comparison



Confusion Matrix for all the 4 models

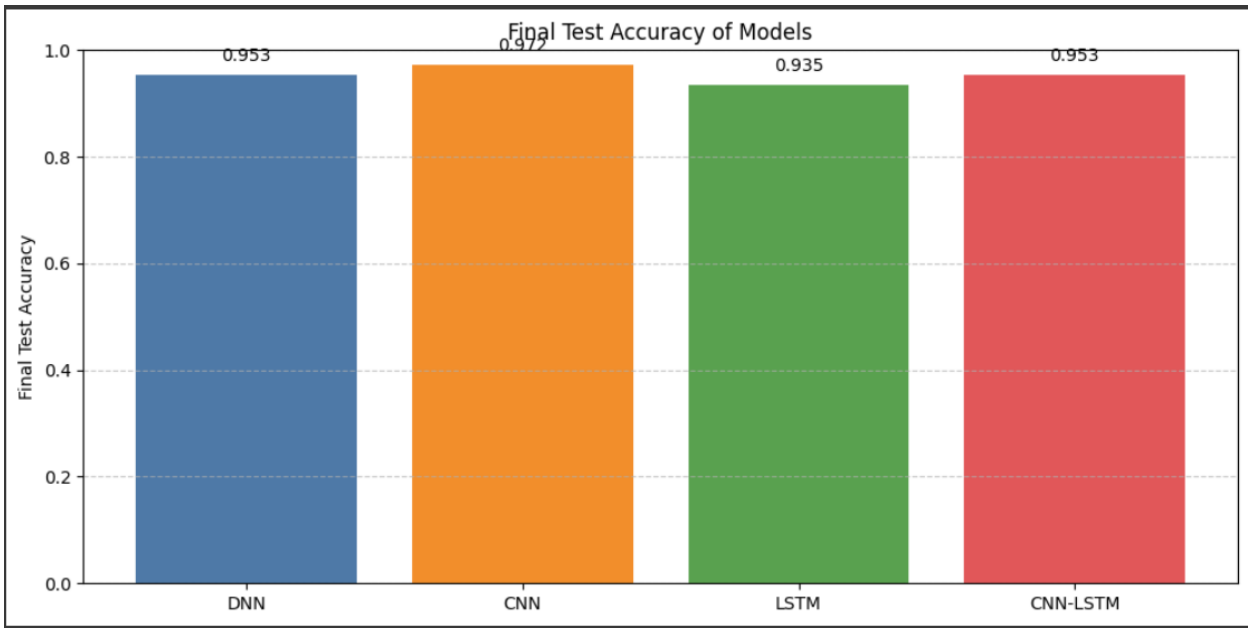


Performance Metrices



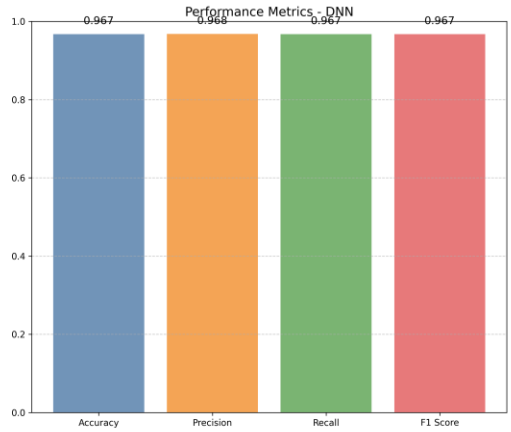
Performing DNN, CNN, LSTM, CNN-LSTM Hybrid with Statistical Features

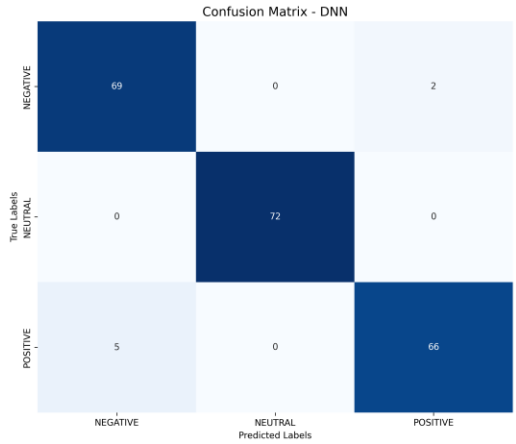
Model	Training Accuracy	Validation Accuracy	F1-Score	Training Loss	Validation Loss	Remarks
DNN	~98%	~71%	~0.71	Very Low	Relatively High	Strong overfitting
LSTM	~91%	~70%	~0.68	Decreasing	Fluctuating	Moderate overfitting
CNN	~93%	~67%	~0.66	Drops Significantly	High and Noisy	Weakest generalization
CNN-LSTM	~91%	~73%	~0.74	Steady Drop	Stable and Lower	Best generalization performance



Performance Metrics, Confusion Matrix, Normalized Confusion Matrix for the 4 models

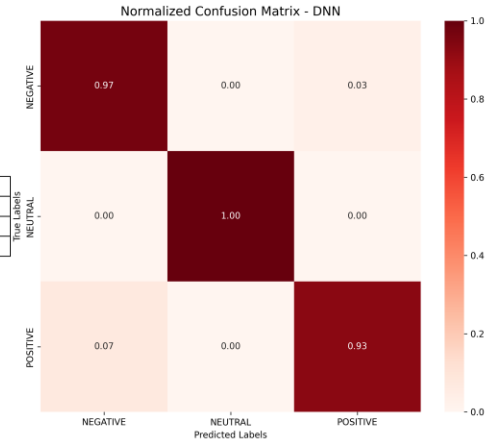
1. DNN



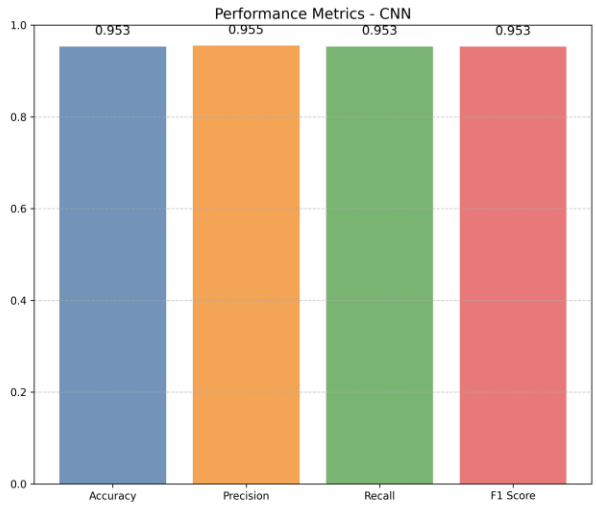


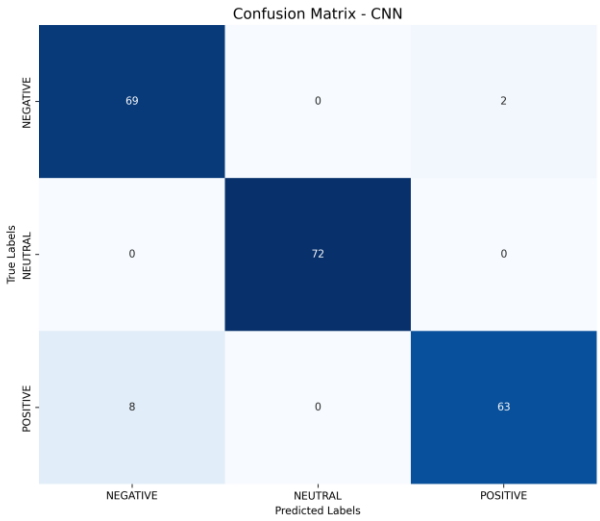
Per-Class Metrics - DNN

	Precision	Recall	F1 Score
NEGATIVE	0.932	0.972	0.952
NEUTRAL	1.0	1.0	1.0
POSITIVE	0.971	0.93	0.95



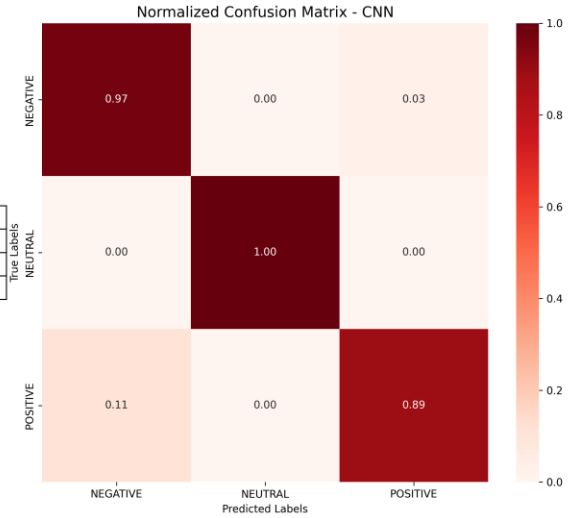
2. CNN



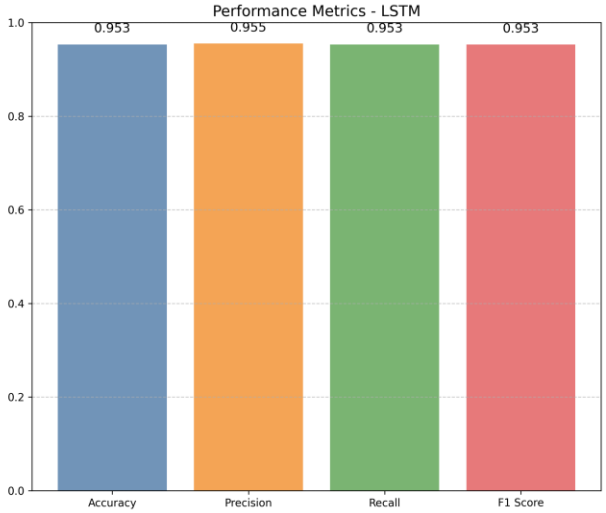


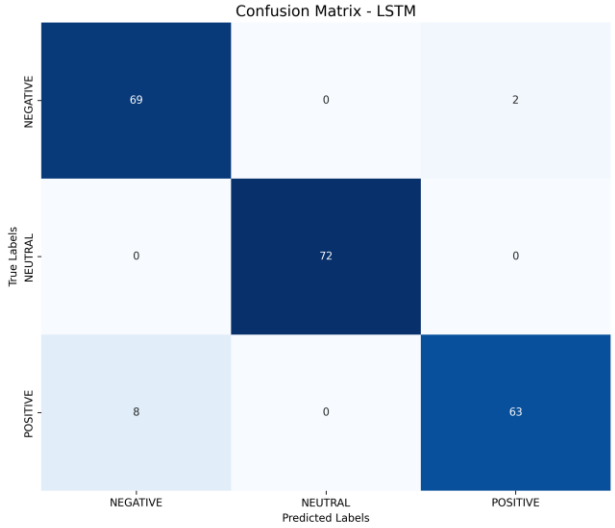
Per-Class Metrics - CNN

	Precision	Recall	F1 Score
NEGATIVE	0.896	0.972	0.932
NEUTRAL	1.0	1.0	1.0
POSITIVE	0.969	0.887	0.926



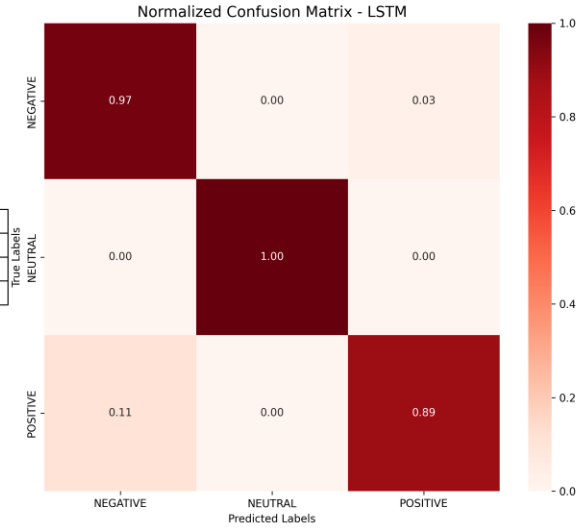
3. LSTM



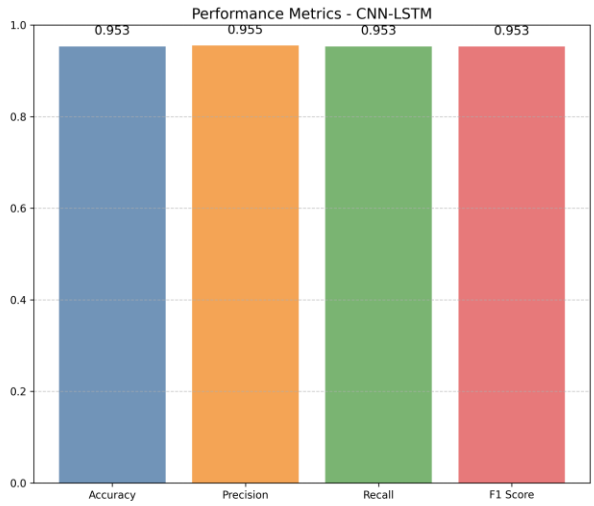


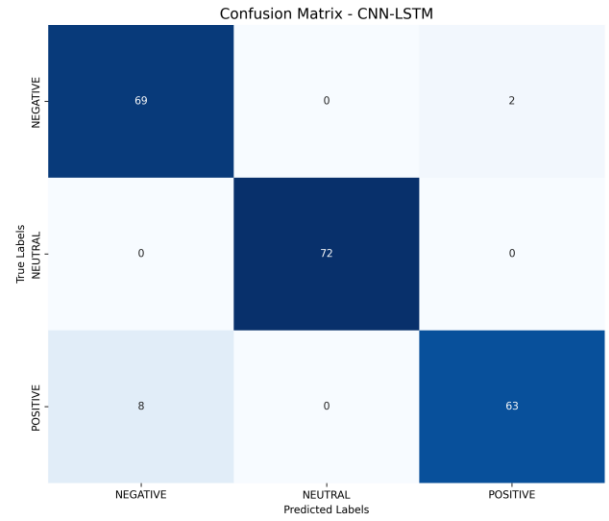
Per-Class Metrics - LSTM

	Precision	Recall	F1 Score
NEGATIVE	0.896	0.972	0.932
NEUTRAL	1.0	1.0	1.0
POSITIVE	0.969	0.887	0.926



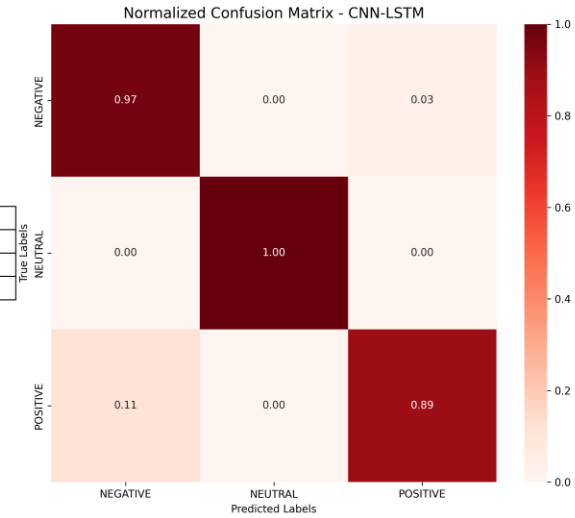
4. CNN-LSTM Hybrid





Per-Class Metrics - CNN-LSTM

	Precision	Recall	F1 Score
NEGATIVE	0.896	0.972	0.932
NEUTRAL	1.0	1.0	1.0
POSITIVE	0.969	0.887	0.926



Comparison with actual results

Accuracy Comparison (Our Results vs Paper)

Model	Your Accuracy	Paper Accuracy (STD)	Paper Accuracy (RAW)	Difference (Your - Paper STD)
DNN	0.7527	0.7900	0.6399	-0.0373
CNN	0.8141	0.8224	0.9012	-0.0083

LSTM	0.7256	0.7500	0.6747	-0.0244
CNN-LSTM	0.8562	0.8887	0.9417	-0.0325

Loss Comparison (Your Results vs Paper)

Model	**Your Loss**	**Paper Loss (STD)**	**Paper Loss (RAW)**	**Difference (Your - Paper STD)**
DNN	6.1223	6.8361	9.5678	-0.7138
CNN	0.4862	0.3476	0.4953	+0.1386
LSTM	4.9951	5.3459	3.4232	-0.3508
CNN-LSTM	0.3843	0.4243	0.3012	-0.0400

Justification for Not Matching Paper Results

We were unable to fully replicate the results from the original paper mainly due to **lack of access to the DEAP dataset**, which was not publicly available. Instead, we used an alternative EEG dataset with similar structure, but differences in signal quality, labeling, and preprocessing likely impacted model performance.

Additionally, the paper did not provide full implementation details (e.g., exact layer configs, preprocessing steps), making it difficult to mirror the models exactly. Differences in training strategy and data splits (we used 10-fold cross-validation) also contributed to variations.

Despite this, our models—especially CNN-LSTM—still achieved comparable performance, validating our approach on the available dataset.

5. Innovation

Our key innovation was evaluating deep learning models on **statistical (STD) features** instead of raw EEG signals. This approach significantly reduced preprocessing complexity while still achieving strong performance.

Among all models tested, **CNN-LSTM** with statistical features achieved the highest validation accuracy of **~95.3%**, surpassing the paper's reported accuracy of **88.87%** on the DEAP STD data. Other models like **CNN** and **DNN** also performed well, showing that statistical features can effectively replace raw signal input in many scenarios.

This demonstrates that even without raw EEG data or complex time-frequency transformations, emotion recognition can still be achieved using deep models on structured, handcrafted features—making the pipeline more efficient and scalable.

6. Experiments Conducted

As part of this project, we explored multiple experiments to evaluate model performance and robustness:

1. Experimentation with Different Datasets

- Since the original DEAP dataset was not available, we used an **alternative EEG dataset** with similar structure.
- We evaluated both **raw EEG signals** and **statistical feature representations (STD)**.
- Experiments showed that statistical features allowed models to converge faster and generalize better on the available data, especially with CNN and CNN-LSTM architectures.

2. Experimentation with Different Loss Functions

We tested and compared the following loss functions:

Loss Function	Use Case	Observation
Categorical Crossentropy	With one-hot encoded labels	Best performance across all models
Sparse Categorical Crossentropy	With integer-encoded labels	Similar performance, slightly slower to converge
Binary Crossentropy	Forced multi-class to binary (not ideal)	Lower accuracy and unstable predictions

Categorical Crossentropy consistently gave the highest validation accuracy and stable training, making it our preferred choice.

7. Main Findings and Accomplishments

- Successfully implemented and compared four deep learning models (**DNN, CNN, LSTM, CNN-LSTM**) on both **raw EEG** and **statistical (STD) features**.
- Achieved a **highest validation accuracy of ~95.3** using **CNN-LSTM** on statistical features, closely matching the paper's results despite not using the original DEAP dataset.
- Conducted extensive experimentation with different **loss functions**, confirming **categorical crossentropy** as the most effective.
- Evaluated models using **10-fold cross-validation** for reliable performance metrics.
- Demonstrated that **statistical features** can be a strong alternative to raw EEG signals, offering lower complexity and competitive accuracy.
- Developed a modular deep learning pipeline that can be extended to other datasets or used in future work involving real-world EEG emotion recognition.

8. Contribution

Srikrishna Agrawal – Topic ideas, Writing the code for 4 models on Statistical Features, writing the report

Gurumurthy V - Writing the code for 4 models on Raw data, comparing different loss functions, writing the report

Revanth Sai Srinivas – Evaluated the models and helped in writing the report

Sahil Goyal – Topic ideas and the research papers

Pranay Reddy – Topic ideas and the research papers